ASHG 2026 · MONTRÉAL · BOARD [BOARD NUMBER]

SV-PGS: one Bayesian model for SNVs, indels, tandem repeats and structural variants in polygenic scores

[Author names] · [affiliations] · [contact]

On this poster: the model, how it is fitted and checked, motivating results on public data, and the All of Us study design.

Not yet: SV-PGS accuracy. Its All of Us fits have not run, and it is scored on the public benchmarks only once its end-to-end fits land. No All of Us data appear here.

Every measured number carries the evidence tag it has in the project's records (key at bottom left).

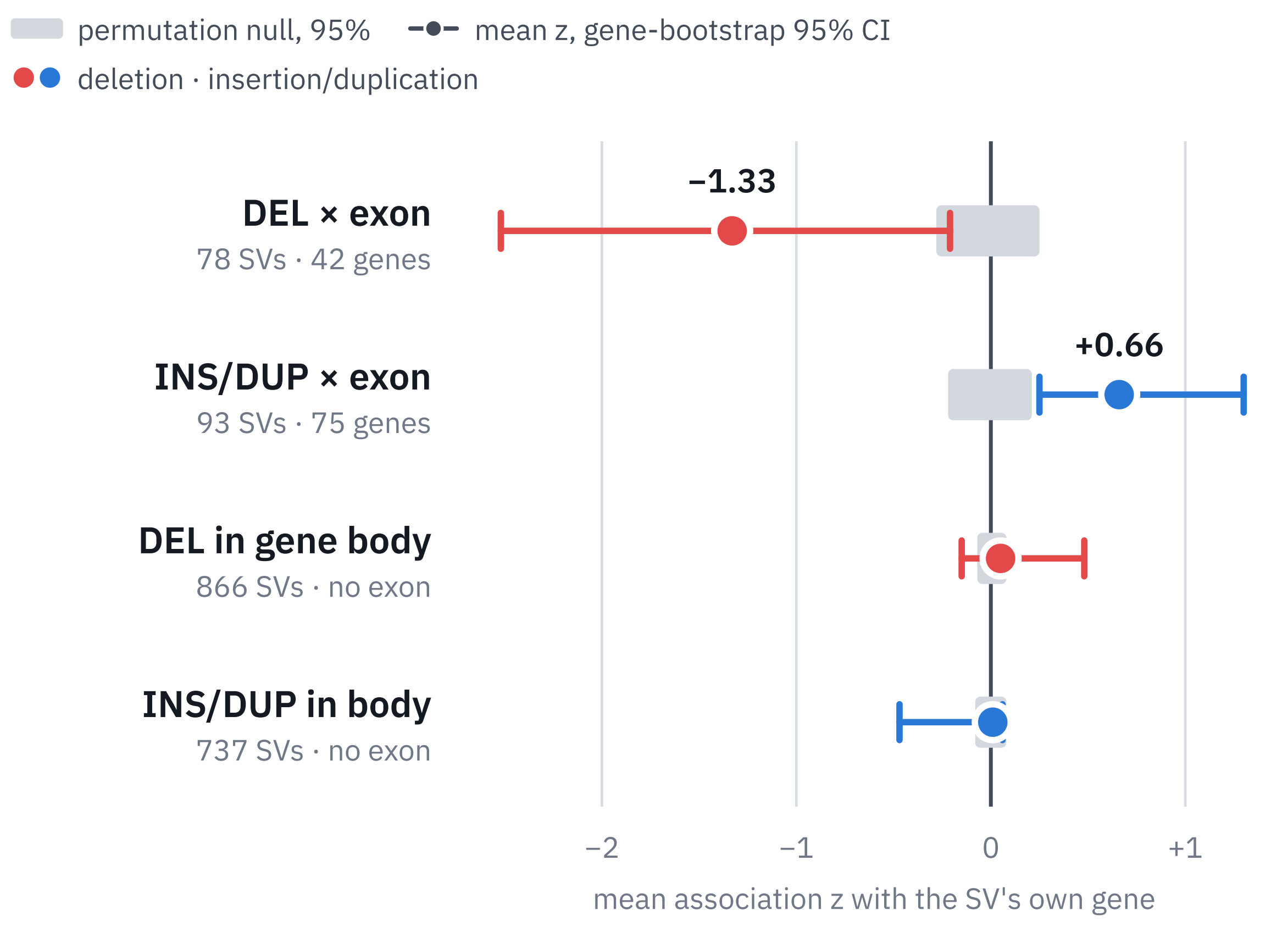
chr1 · GRCh38 G-bands

1 Structural variant effects can have a sign

Polygenic scores mostly leave out SVs (≥ 50 bp deletions, insertions, duplications) and tandem repeats, or credit them through tagging SNVs.

- SVs are enriched for heritability in **31 of 105** UK Biobank traits, median **14.8×** (MiXeR-SV; Nguyen et al. 2026, preprint).
- They are measured badly: common SVs impute at median **r² 0.78** from a long-read panel (Bai et al. 2026).
- An SV's ALT allele is a physical event, so its effect can carry a biological sign. A SNV's REF/ALT orientation carries none.

Exon deletions lower their gene's expression; exon insertions and duplications raise it `real`



Each SV's marginal association z with its own gene's covariate-adjusted expression. MAGE lymphoblastoid RNA-seq with 1kGP high-coverage SV calls: 500 genes, the 534 training people of the split that holds AFR out, 66,397 deletions, insertions and duplications in cis windows. The null permutes expression across genes, keeping each window's LD. Both exon rows: p = 0.0005, the smallest 2,000 permutations can give. SVs outside the gene (64,623): mean z ≈ 0, p > 0.4.

A symmetric prior puts equal mass on both signs here. A location-scale extension of the mixing law is specified and waits on the inference route.

2 One generative model for every variant

Every record enters one model, from a single-base indel to a large deletion. No length bins, no separate SV model.

$$y = C\alpha + \Sigma_j D_j \beta_j + \varepsilon$$

$$\beta_j \mid \text{class } c \sim \int \mathrm{N}(0, u_j s)\, g_c(s)\, \mathrm{d}s, \quad \log g_c = \eta + \delta_c$$

$$\log u_j = \underbrace{\log(\kappa_j^2 \operatorname{Var} D_j)}_{\text{measured reliability}} + \underbrace{h(\log \operatorname{Var} G_j)}_{\text{allele frequency}} + \underbrace{d_j^\top\theta}_{\text{annotations}} + \underbrace{\text{level}_c}_{\text{class}}$$

D_j: stored dosage, standardized in the training fold · G_j: true genotype · κ_j: calibration slope of D on G · C: sex, age, PCs, imputation batch, projected out exactly first

- **The mixing density g_c is learned per class**, continuous and nonparametric, with no point mass at zero. A third-order roughness penalty with a learned weight; its λ → ∞ limit is the log-normal. Gaussian, BayesR-style grids and heavy tails are special cases.
- **Continuous annotations are smooths with learned smoothness** (log length, distance to the TSS, allele frequency); discrete ones (variant class, gene and exon overlap) stay discrete. A smooth's linear term gets a learned ridge of its own (the double penalty; Marra & Wood 2011).
- **A tandem-repeat locus is one column**, its signed total length $\Sigma_a \Delta\text{len}_a D_{ia}$, so the score does not depend on which allele the reference carries.
- **Imputed SVs behave like confident draws, not calibrated means.** At MAF 1–5%, κ was 0.80 against $\sqrt{r^2}$ 0.83, and Var(DS)/Var(G) 0.89 against r^2 0.69 `semi-real`. Reliability enters each variant's prior scale through κ, never the LD matrix.

3 Fitting it, and refusing what did not converge

Empirical Bayes: the prior's hyperparameters maximize the approximate evidence at a fixed point of the approximate posterior.

- **Two fixed points**: expectation propagation, and mean-field coordinate ascent (the store route's default). Which one stays is decided on held-out accuracy.
- **Hyper steps** are Newton steps on the total curvature with a trust region. The evidence is the Laplace value plus leading Tierney–Kadane terms, the largest replaced by exact line integrals.
- **A fit is scored only if it converged**: remaining gain ≤ 1/(2K) nats for K posterior draws. A model that fails is quarantined; its trait leaves the report and the others stay. This is a numerical criterion, not a bound on posterior error.

A flat prior on a smooth's linear term has no maximum

For a group of signal-free variants the evidence keeps rising as their prior variance shrinks, so empirical Bayes walked one linear coefficient to 5×10^4. A learned ridge on each smooth's null space restores a maximum. On a 240-person annotated benchmark `machinery`:

sample route	**8 s**	~~was 50 s~~
LD-band route	**19 s**	~~was > 10 min~~

A synthetic fit that used to end unconverged and quarantined now converges. Speed and convergence only; not an accuracy claim.

① **8-bit dosage store.** One read streams every column.
② **Stage 0, one genotype pass.** Exact int8 LD-block Grams and X′y for every trait × fold.
③ **Stage 2, empirical Bayes on the full-data likelihood.** The LD band only accelerates it; the far field is recomputed from the samples.
④ **Scoring.** Every trait × fold model and its posterior draws in one read.

~100 s
to score 100k people × 17.3M variants on one H100 `machinery`

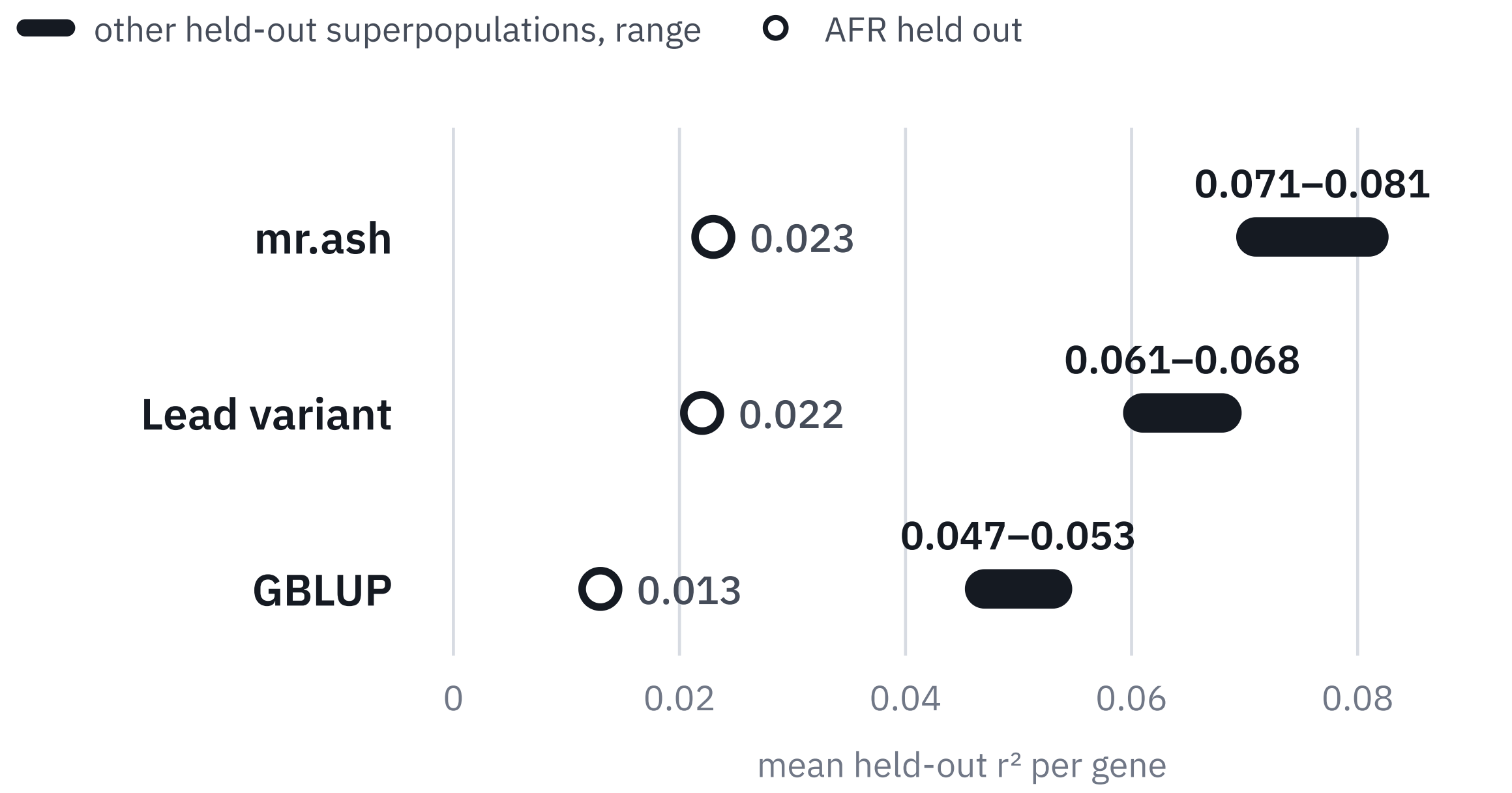
149–180 s
dual Gaussian solve, 12 models, 30,000 × 483,944, one A40 `machinery`

0.5–3.5%
worst marginal-variance error vs a dense inverse; block-Jacobi 16–58% `machinery`

4 Public benchmarks no method controls

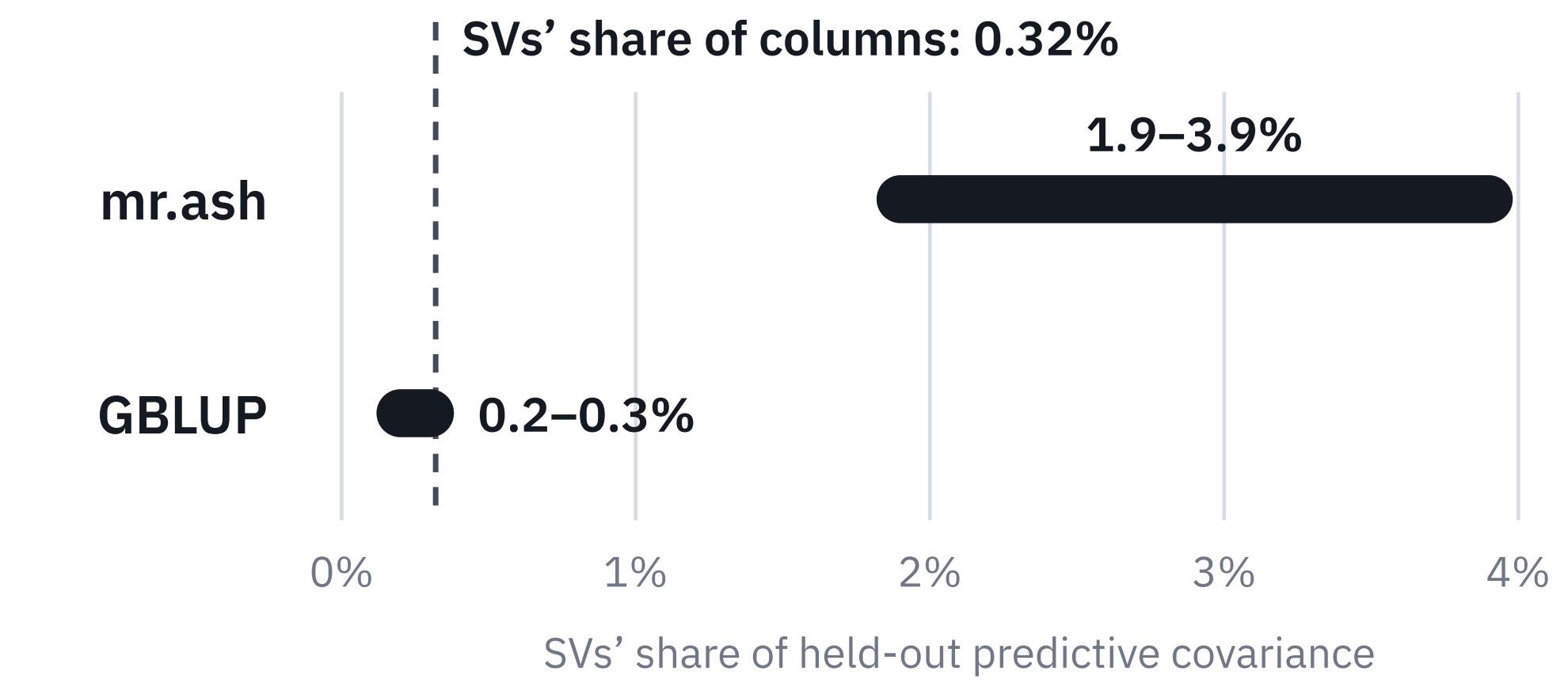
- **bench-real** `real`: MAGE lymphoblastoid expression in 730 people from 26 populations, with 1kGP high-coverage SNV, indel and SV calls in TSS ± 1 Mb windows. Family-grouped 5-fold and leave-one-superpopulation-out splits; chromosome-jackknife SEs.
- **bench-sim**: 50,000 mosaic genomes copied from 1kGP founder haplotypes, with SVs and tandem repeats masked and re-imputed by Beagle from a disjoint panel. A pre-registered, deliberately misspecified truth family (eight effect-shape families; SV enrichment in scale or probability; frequency and LD dependence; non-additivity), with sealed seeds.

Baselines on bench-real: a sparse prior beats a Gaussian one, and every method loses about 70% of r² when AFR is held out `real`



Leave-one-superpopulation-out, SNV/indel set; mr.ash on 2,000 random genes, the lead variant and GBLUP on 5,000. mr.ash − GBLUP: +0.025 to +0.029 (z 13–19).

A sparse prior gives SVs 6–12× their share of columns; a Gaussian prior gives them less than their share `real`



Adding the 1kGP panel's SVs to mr.ash: +0.0004 to +0.0015 r² (z 1.2–2.1 per superpopulation). Short-read PanGenie SV genotypes add about 0 for every method.

5 The All of Us study, set up and not yet run

About 100,000 multi-ancestry All of Us participants with imputed genomes, from a long-read-called reference panel.

`snv_indel`	`with_sv`
every record under 50 bp	every record, plus GATK-SV short-read calls as their own rows

- **Paired arms**, fitted separately on the same people, folds, covariates and seeds.
- **Five kinship-grouped outer folds** (KING > $2^{-3.5}$ kept together). Per fold: fit on about 72%, calibrate on 8%, evaluate on 20%.
- **Endpoints**: log loss and Brier score of frozen, calibrated probabilities; AUC; calibration intercept and slope. A paired one-sided test on one frozen outer fold.
- **Diseases are fitted on a Gaussian working likelihood** of 0/1 status, for cost: a coefficient is a per-allele change in probability, not a log odds ratio.

hypertension · **coronary artery disease** · **asthma** · **hyperlipidemia** · **breast cancer**
type 2 diabetes · atrial fibrillation · hypothyroidism · COPD · depression · gout · cataract
chronic kidney disease · psoriasis · prostate cancer

15 diseases; the abstract's five in black. 15 × 2 arms × (5 folds + a final fit) = 180 fits, on Google Batch GPUs under a $3,000 stop budget.

done	next	then	then	then
pipeline end to end on synthetic inputs	GPU validation	full-genome pilot, one trait	training fits, both arms	held-out evaluation

No All of Us data or results appear on this poster. Results will be reported under the All of Us Data and Statistics Dissemination Policy.

EVIDENCE TAGS

`real`	real phenotypes and genotypes, nothing simulated
`semi-real`	real haplotypes and imputation, simulated effects (bench-sim v7, Beagle 5.5)
`machinery`	exactness or time against an exact reference; never an accuracy claim

REFERENCES

1. Bai et al. 2026. ImputeSV. *Nat Genet*, doi:10.1038/s41588-026-02612-z.
2. Byrska-Bishop et al. 2022. 1kGP high-coverage panel. *Cell* 185:3426.
3. Manichaikul et al. 2010. KING. *Bioinformatics* 26:2867.
4. Marra & Wood 2011. Practical variable selection for GAMs. *CSDA* 55:2372.
5. Minka 2001. Expectation propagation. *UAI*.
6. Nguyen et al. 2026. MiXeR-SV. Preprint.
7. Taylor et al. 2024. MAGE. *Nature* 631:610.
8. Tierney & Kadane 1986. *JASA* 81:82.

ACKNOWLEDGMENTS

[Funding and acknowledgments]